

# MicroStrategy Bitcoin Mispricing

## Thesis

MicroStrategy (MSTR) holds about 3.8% of the world's Bitcoin supply, and its market cap is primarily based on Bitcoin prices. Using this information, we can easily calculate whether MSTR stock is overpriced or underpriced relative to the amount of Bitcoin they are holding.

## Background

As of April 2026, MSTR holds 761,068 bitcoin, valued at around \$56.9 billion. By taking historical dates of when MSTR updated their bitcoin holdings, we can calculate an implied price of MSTR stock, and compare it to the actual price to see if there is a significant mispricing.

Throughout this pitch, the formula for calculating the implied MSTR price is:

Implied Price = (BTC Held / Shares Outstanding) \* BTC Price

This formula shows us how much MSTR stock should be worth given current Bitcoin prices and its holdings relative to its current price.

MSTR announces changes to their Bitcoin holdings usually after markets close, so I shifted the dates at which they announced updated holdings to 1 day later to avoid look-ahead bias.

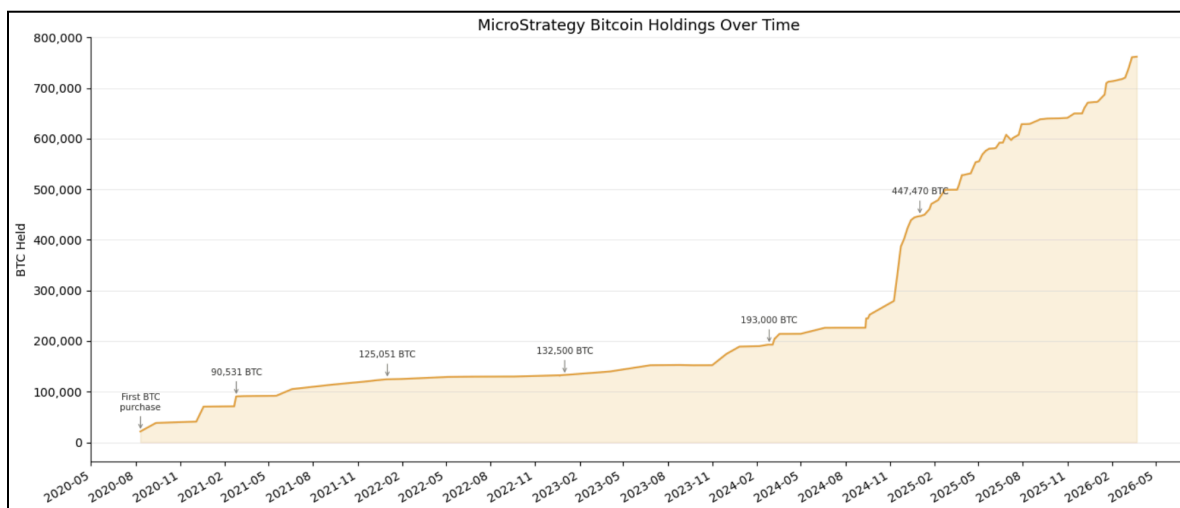


Figure 1: MSTR Bitcoin Holdings Over Time.

## EDA

Firstly, I wanted to see if my implied price formula actually accurately calculates a fair implied price for MSTR. I can do this by running an ADF test to see whether the actual prices are mean-reverting back to my implied price.



Figure 2: ADF Test on Formula for Implied MSTR Price.

It looks like MSTR trades at a pretty consistent 1.03x premium over our implied price close to 1-to-1, as my model would initially suggest.

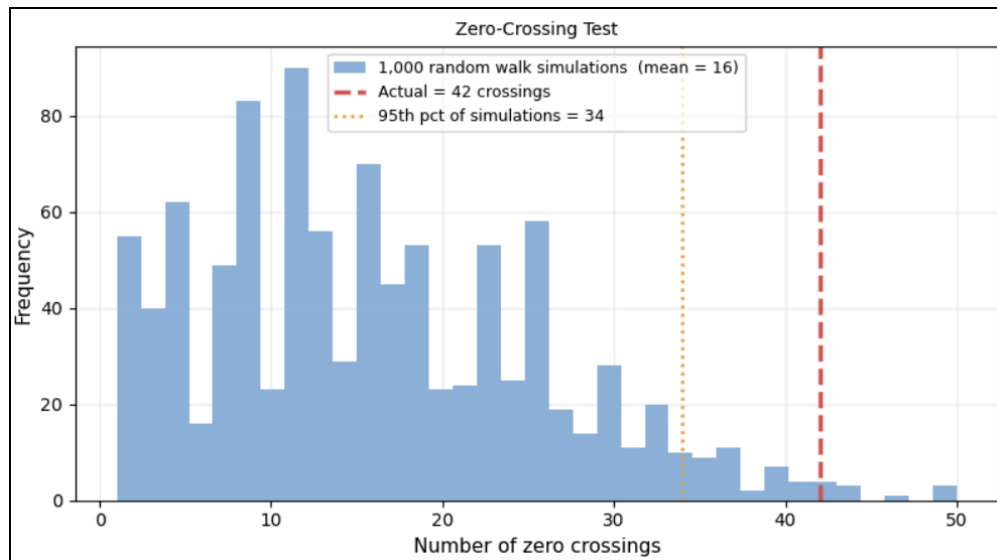
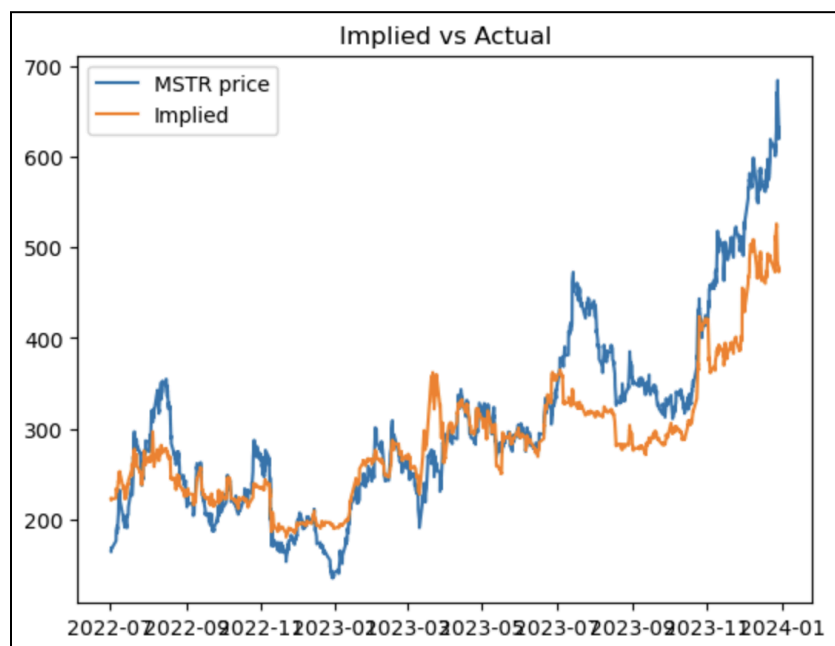


Figure 3: Zero-Crossing Test vs Random Walks

Nevertheless, our formula seems to work really well, with it accurately pricing MSTR stock prices based on our ADF test.



```
Test statistic: -3.6681
p-value:      0.0201
Critical values:
  1%: -3.9013
  5%: -3.3388
 10%: -3.0463

Cointegrated: YES ✓
```

Figure 3: Implied Price Series vs. Actual Price Series & Cointegration Test

To improve robustness, I also ran a Cointegration Test to see whether my implied price formula really moves with MSTR prices.

Trade Expression:

- Use a rolling mean Actual / Implied using x lookback hours
- When rolling z-scores exceed the rolling average by a certain threshold, we long/short MSTR, believing it to be unfairly priced relative to its Bitcoin value
- To hedge out crypto directional risk, we take an opposing position on Bitcoin. Our hedge weights are based on ["btc\_per\_share"]
- We exit positions when Actual / Implied returns back to the mean

Example Trade:

- Actual / Implied = 1.30
- Rolling Mean = 1.0
- MSTR shares are trading at a 30% premium relative to their Bitcoin value
- I go short on 100 MSTR shares, go long on Bitcoin
- Hedge =  $1,000 \times 0.00023 = 0.23$  BTC long

I wanted to see if the correlation breaks when Bitcoin has a lot of momentum. If there is some sort of correlation between Bitcoin's momentum and Actual / Implied, that means our model will enter bad trades when Bitcoin is gaining a lot of upside/downside momentum, telling us that Actual / Implied is too high even though it isn't, depending on the momentum of Bitcoin.

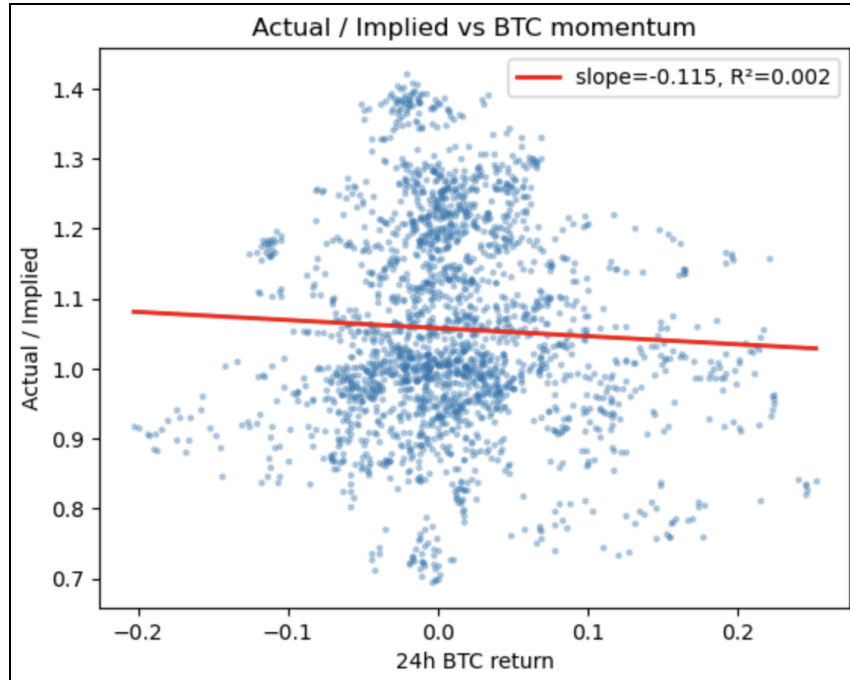


Figure 4: Bitcoin Momentum vs Actual / Implied

There does not seem to be a significant correlation between Bitcoin's momentum and Actual / Implied, which simplifies our trade expression. The deviations we are trading are driven by MSTR-specific noise, not by Bitcoin's direction. We can therefore treat our mNAV signal as independent of Bitcoin momentum and do not need to filter trades based on how much Bitcoin has moved.

I believe using a rolling mean is a much better way to go about this trade idea rather than sticking with a strict mean like 1.11x, as crypto markets and accessibility to crypto are rapidly expanding, as well as MSTR and its actual software company.

I can prove this by the plot below. There appears to be a structural shifting of Implied / Actual. As we near the end of 2024, we see different regimes where the MSTR Price / Implied price is different. By making a slow rolling mean, we can compensate for the changing regimes.

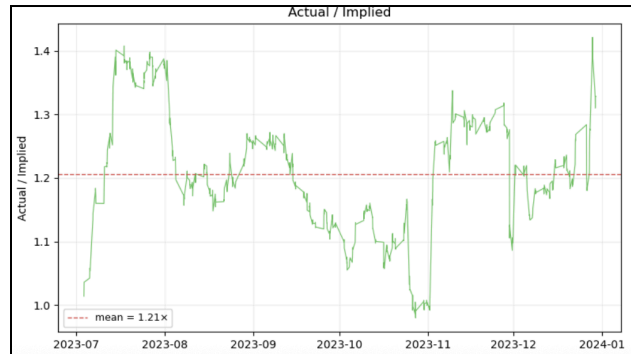
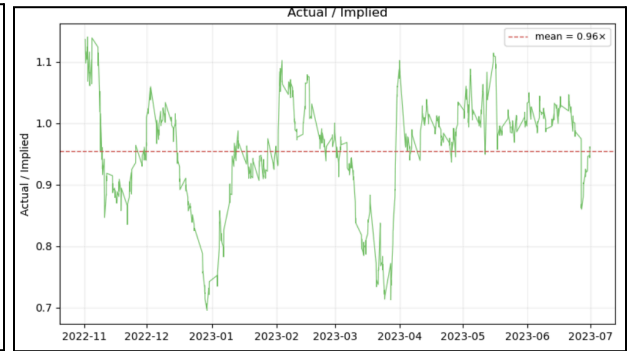
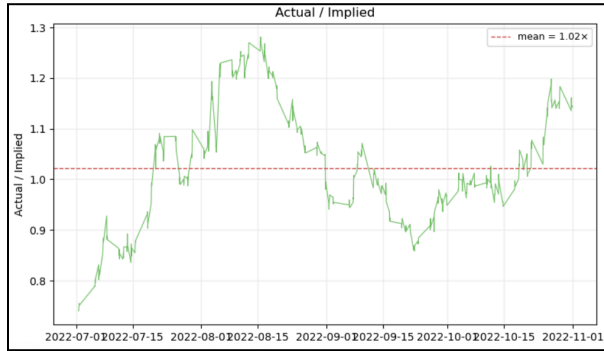
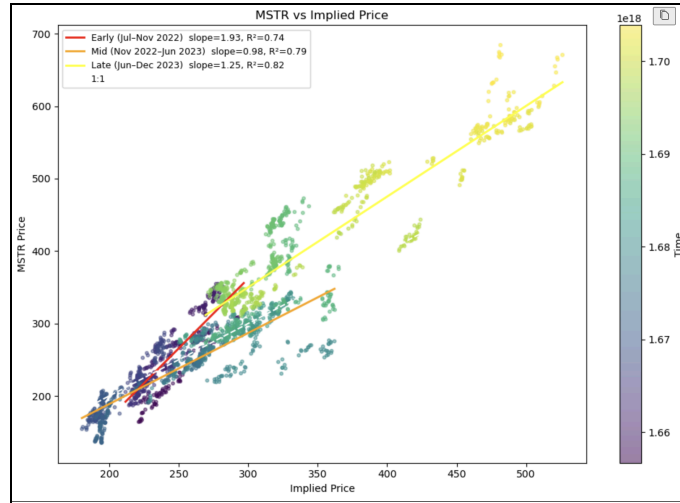


Figure 5: Different Actual / Implied Regimes

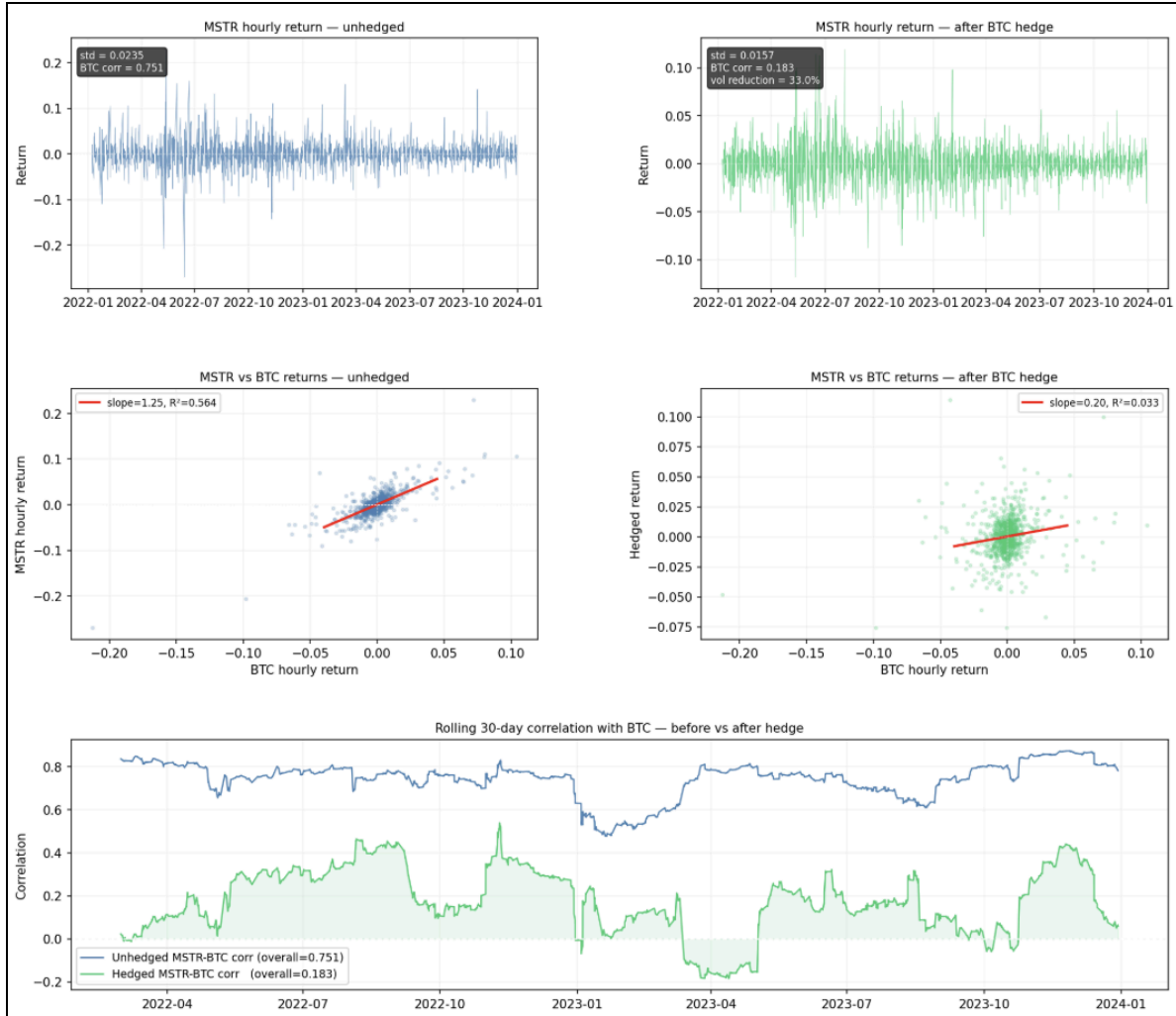


Figure 6: Hedging Out Bitcoin Exposure Using MSTR's Bitcoin Per Share

In my trade expression, I wanted to hedge out Bitcoin risk as much as possible. I want my correlation in MSTR returns to be as close to 0 with Bitcoin, which would mean I hedged out Bitcoin risk entirely. To hedge out crypto directional risk, we take an opposing position on Bitcoin.

$$\text{hedged\_ret} = \text{mstr\_ret} - \text{hedge\_ratio} \times \text{btc\_ret}$$

$$\text{hedge\_ratio} = (\text{btc\_per\_share} \times \text{btc\_price}) / \text{mstr\_price}$$

There is still a little directional risk to Bitcoin using this hedge method. This is because MSTR moves more like a leveraged Bitcoin rather than just a 1-to-1 movement. I can fix this by running an OLS to find optimal hedge weights rather than deriving hedge weights from the Bitcoin per share.

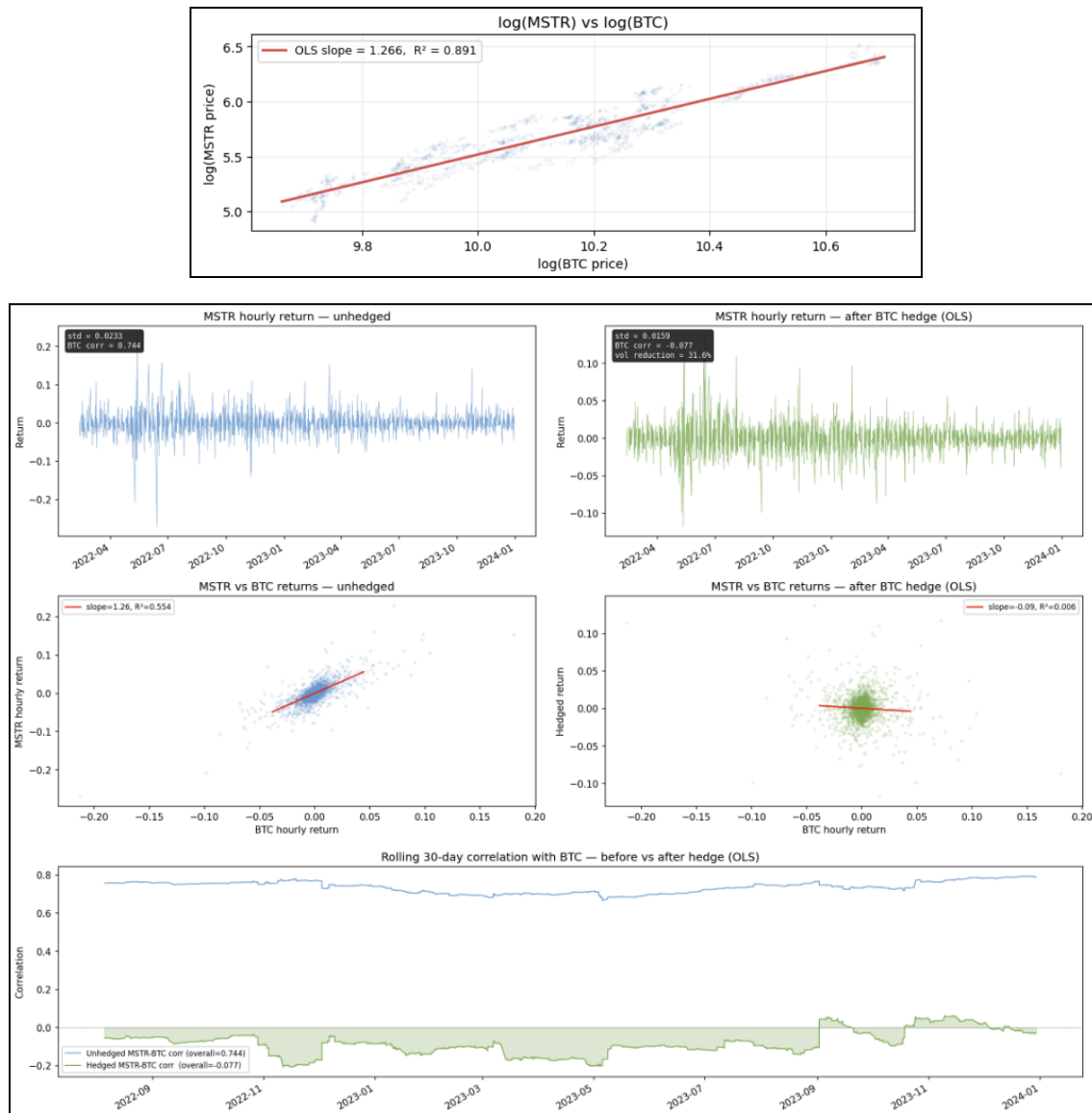


Figure 7: Hedging Out Bitcoin Exposure using OLS

Now the slope is more towards 0, hedging out Bitcoin exposure completely.

## Optimization

Since MSTR is pretty liquid, I will implement a transaction cost of 30 bps.

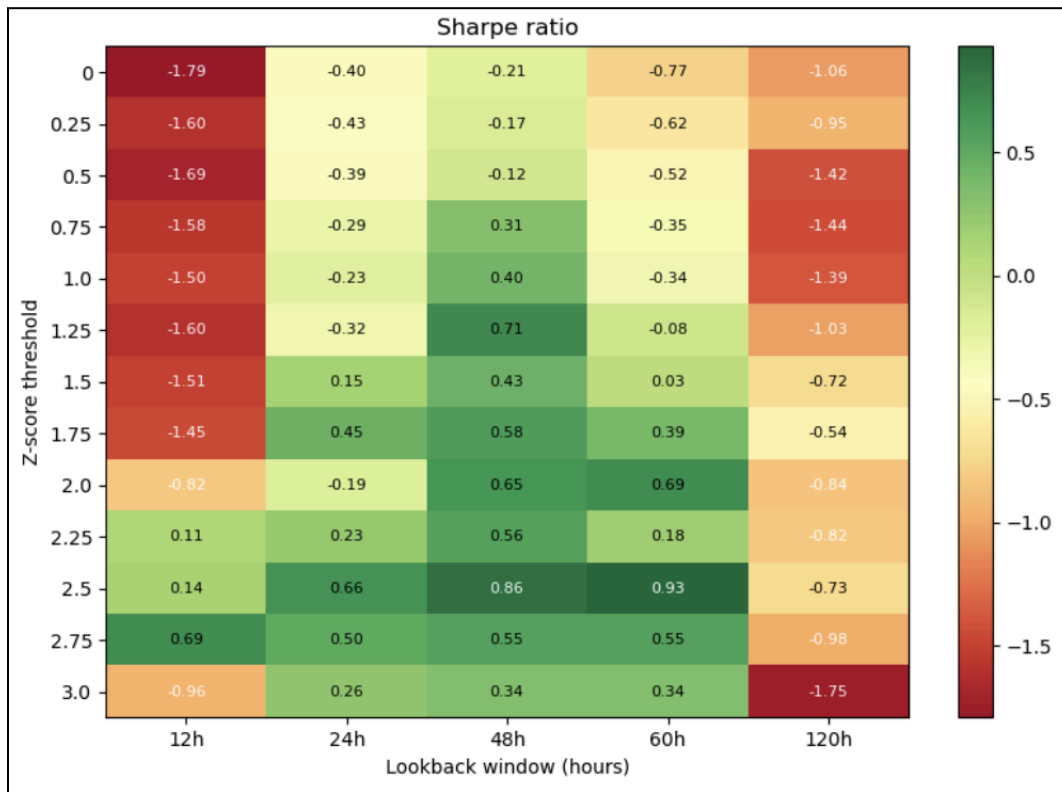
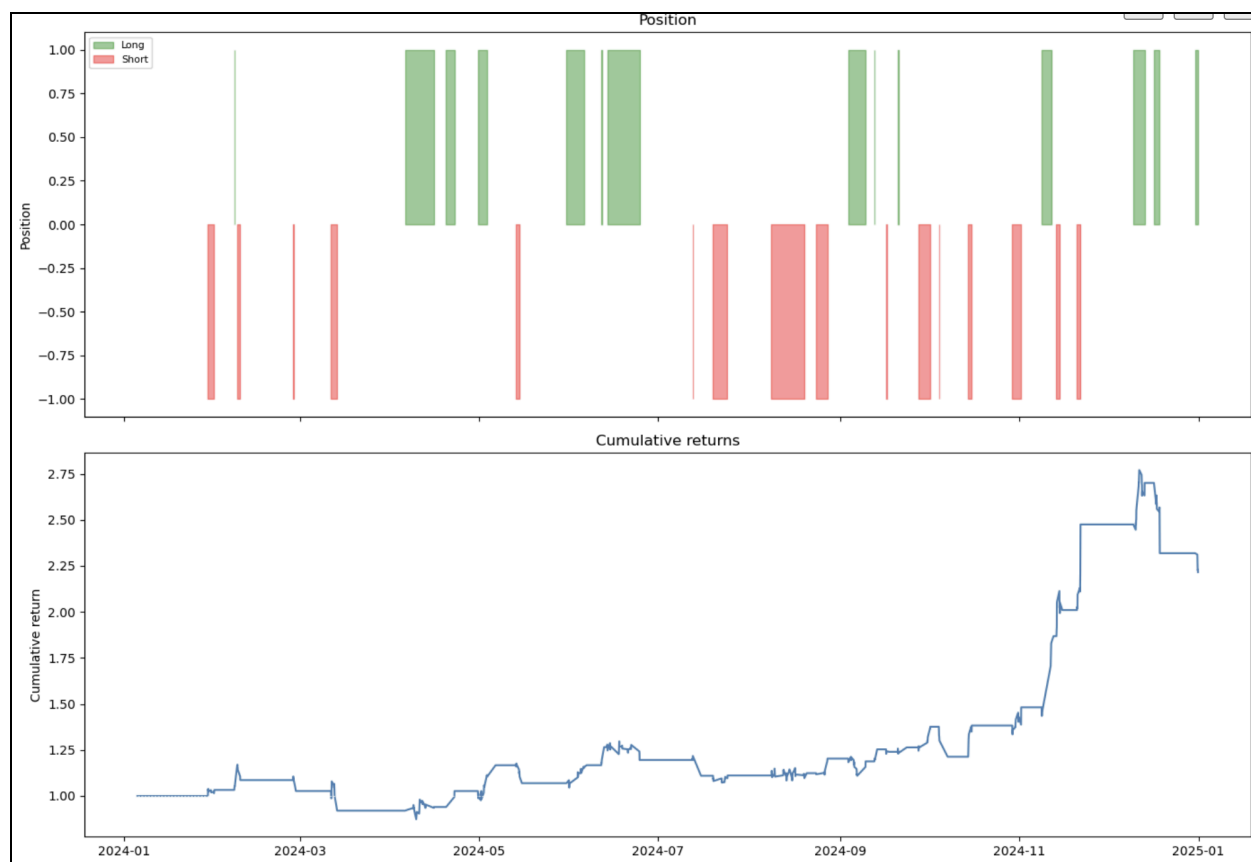


Figure 8: Heatmap of Optimal Lookback Window and Z-Score Threshold.

I chose to pick the parameters of a 48-hour lookback and a 2.5 z-score threshold. I ignored the high sharpes to the left since the number of trades throughout the year was less than 10.



```

---
Number of trades:      30
Win rate:              60.0%
Avg P&L per trade:     2.68%
Avg winning trade:     10.07%
Avg losing trade:      -8.41%
Avg duration (hours):  85.4
---
Long trades:   14  (win rate: 64.3%)
Short trades:  16  (win rate: 56.2%)

```

Figure 9: PnL Graph of Heatmap Parameters.

It looks like I made a few huge wins around November - December. I wanted to look into the cause behind that.

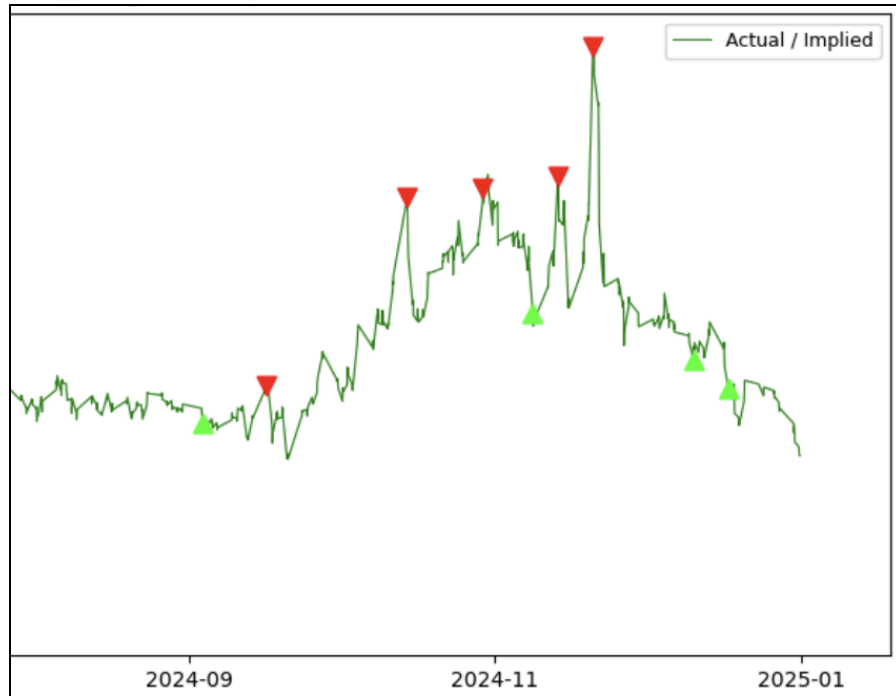


Figure 10: Dates of Extreme Deviation

There are a few large spikes in Actual / Implied during this time period.

November 5–11 2024: Donald Trump's election victory, funded heavily by the crypto industry, propelled the stock significantly. MSTR shares climbed 60% in the weeks following the November 5 election and finally exceeded their dot-com era high on November 11.

Late November / December: Saylor announced that over a six-day stretch starting December 9, the company acquired 15,350 bitcoins for \$1.5 billion. Each aggressive BTC purchase announcement temporarily sent MSTR into premium territory.

## Backtest



```
Strategy return:      28.1%
Strategy Sharpe:     0.91
---
Number of trades:    17
Win rate:            58.8%
Avg P&L per trade:   1.93%
Avg winning trade:   9.32%
Avg losing trade:    -8.63%
Avg duration (hours): 142.1
---
Long trades:      11  (win rate: 54.5%)
Short trades:     6  (win rate: 66.7%)
```

Figure 11: PnL Graph of Backtest